

Distributed-Parameter Modelling of a Three-Phase Overhead Transmission Line and Phase-Current Unbalance as a Function of Length and Apparent Power

Litgrid

Abstract

This note documents the analytical model used by the Litgrid application to compute voltage and current profiles along a three-phase overhead transmission line. Per-unit-length matrices $\mathbf{R}, \mathbf{L}, \mathbf{C}, \mathbf{G}$ are obtained from tower geometry by decomposing each conductor's self impedance into an internal (skin) contribution and an external (geometric) contribution, building a fully-coupled cable-to-cable matrix, reducing it first across the bundled conductors of a phase, and finally Kron-eliminating the overhead ground wires. The line is then solved as a distributed-parameter system using the matrix exponential of its state-space generator. A two-point boundary problem is posed with a symmetrical positive-sequence voltage at one terminal and a specified three-phase complex power $S = P + jQ$ at the same (or opposite) terminal. The natural asymmetry of the per-unit-length matrices for non-transposed geometries drives the phase-current unbalance, which is quantified and presented as a function of length and apparent power. A complete worked example for a 110 kV single-circuit line closes the document.

1 Notation and Conventions

Bold symbols denote vectors and matrices in the conductor- or phase-domain. Phasors are RMS, with the cosine reference. The phase rotation operator is $a = e^{j2\pi/3}$ and the positive-sequence unit vector is

$$\mathbf{u} = \begin{bmatrix} 1 \\ a^2 \\ a \end{bmatrix}. \quad (1)$$

Angular frequency $\omega = 2\pi f$. Distances along the line are in kilometres, so $\mathbf{R}$ is in Ω/km , $\mathbf{L}$ in mH/km , $\mathbf{C}$ in nF/km , and $\mathbf{G}$ in S/km .

2 Per-Unit-Length Parameters of the Coupled Conductor System

The tower carries N conductors in total, comprising the bundled sub-conductors of $n_p = 3$ phase bundles plus $n_g \geq 0$ overhead ground wires. Each conductor i is described by its DC resistance $r_{\text{dc},i}$ at the operating temperature, its outer radius r_i , and its Cartesian position (x_i, y_i) with $y_i > 0$ above the perfectly conducting earth plane.

2.1 Self and Mutual Inductances

Following the classical decomposition of the magnetic flux linking a thin round non-magnetic conductor at the power frequency, the self inductance per unit length of conductor i is the sum of two contributions:

Internal inductance. The flux that resides *inside* the conductor (uniform-current low-frequency approximation):

$$L_{\text{int}} = \frac{\mu_0}{8\pi} \quad [\text{H/m}]. \quad (2)$$

This is the same for every solid round conductor and does not depend on its radius.

Geometric (external) self inductance. The flux that links the conductor through the region between its surface (radius r_i) and the earth-return reference distance D_e below the soil:

$$L_{\text{self},i}^{\text{geom}} = \frac{\mu_0}{2\pi} \ln \frac{D_e}{r_i} \quad [\text{H/m}]. \quad (3)$$

Geometric mutual inductance. Between two distinct conductors i and j separated by the (real) distance $d_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$:

$$L_{\text{mut},ij}^{\text{geom}} = \frac{\mu_0}{2\pi} \ln \frac{D_e}{d_{ij}} \quad [\text{H/m}]. \quad (4)$$

The earth-return depth and the earth-return resistance follow Carson's low-frequency homogeneous-soil approximation:

$$D_e = 658.5 \sqrt{\rho/f} \quad [\text{m}], \quad R_e = \frac{\omega \mu_0}{8\pi} \quad [\Omega/\text{m}], \quad (5)$$

where ρ is the soil resistivity in $\Omega \cdot \text{m}$.

2.2 Series Impedance Matrix of the Coupled System

Assembling (2)–(4) into the per-unit-length series impedance gives, in Ω/km ,

$$Z_{ii} = (r_{\text{dc},i} + R_e) + j\omega (L_{\text{int}} + L_{\text{self},i}^{\text{geom}}) \cdot 10^3, \quad (6)$$

$$Z_{ij} = R_e + j\omega L_{\text{mut},ij}^{\text{geom}} \cdot 10^3, \quad i \neq j. \quad (7)$$

The $N \times N$ symmetric matrix $\mathbf{Z}_{\text{full}}$ collects every cable-to-cable coupling (phase sub-conductors and ground wires alike).

2.3 Potential Coefficient Matrix of the Coupled System

With the earth treated as a perfect equipotential, image-charge theory yields the $N \times N$ potential-coefficient matrix $\mathbf{P}_{\text{full}}$:

$$P_{ii} = \frac{1}{2\pi\epsilon_0} \ln \frac{2y_i}{r_i}, \quad (8)$$

$$P_{ij} = \frac{1}{2\pi\epsilon_0} \ln \frac{D_{ij}}{d_{ij}}, \quad i \neq j, \quad (9)$$

where $D_{ij} = \sqrt{(x_i - x_j)^2 + (y_i + y_j)^2}$ is the distance from conductor i to the *image* of conductor j below the soil. $\mathbf{P}_{\text{full}}$ relates charge to voltage: $\mathbf{V}_{\text{full}} = \mathbf{P}_{\text{full}} \mathbf{Q}_{\text{full}}$.

3 Reduction to the Phase Domain

The N -conductor coupled matrices contain information that is not directly useful for line operation: sub-conductors of a bundle are paralleled, and the overhead ground wires are short-circuited to earth. Two successive linear reductions eliminate these degrees of freedom.

3.1 Bundle Reduction

Let the N physical conductors be partitioned into n_p phase bundles plus n_g ground wires (each ground wire is its own “bundle of size one”). Introduce the connection matrix $\mathbf{T} \in \{0, 1\}^{N \times (n_p + n_g)}$ with $T_{ik} = 1$ if conductor i belongs to bundle k , zero otherwise. The two physical constraints of a bundle are

$$\mathbf{V}_{\text{full}} = \mathbf{T} \mathbf{V}_b, \quad \mathbf{I}_b = \mathbf{T}^\top \mathbf{I}_{\text{full}}, \quad \mathbf{Q}_b = \mathbf{T}^\top \mathbf{Q}_{\text{full}}, \quad (10)$$

expressing equality of potential within a bundle and Kirchhoff summation of the per-conductor currents and charges.

Substituting in $\mathbf{V} = \mathbf{Z}_{\text{full}} \mathbf{I}$ and $\mathbf{V} = \mathbf{P}_{\text{full}} \mathbf{Q}$ and using $\mathbf{Y}_{\text{full}} = \mathbf{Z}_{\text{full}}^{-1}$ yields the bundle-reduced matrices

$$\mathbf{Z}_b = (\mathbf{T}^\top \mathbf{Z}_{\text{full}}^{-1} \mathbf{T})^{-1}, \quad (11)$$

$$\mathbf{P}_b = (\mathbf{T}^\top \mathbf{P}_{\text{full}}^{-1} \mathbf{T})^{-1}. \quad (12)$$

Both have size $(n_p + n_g) \times (n_p + n_g)$. When every bundle is a single conductor, $\mathbf{T} = \mathbf{I}_N$ and the bundle step is the identity.

3.2 Kron Elimination of the Ground Wires

Partition the bundle-reduced matrices into a phase block (subscript p , size $n_p = 3$) and a ground-wire block (subscript g , size n_g):

$$\mathbf{Z}_b = \begin{bmatrix} \mathbf{Z}_{pp} & \mathbf{Z}_{pg} \\ \mathbf{Z}_{gp} & \mathbf{Z}_{gg} \end{bmatrix}, \quad \mathbf{P}_b = \begin{bmatrix} \mathbf{P}_{pp} & \mathbf{P}_{pg} \\ \mathbf{P}_{gp} & \mathbf{P}_{gg} \end{bmatrix}. \quad (13)$$

Imposing $\mathbf{V}_g = \mathbf{0}$ on the ground wires (perfectly bonded to earth) removes the ground-block degrees of freedom and gives the 3×3 phase-domain matrices

$$\mathbf{Z}_{\text{ph}} = \mathbf{Z}_{pp} - \mathbf{Z}_{pg} \mathbf{Z}_{gg}^{-1} \mathbf{Z}_{gp}, \quad (14)$$

$$\mathbf{P}_{\text{ph}} = \mathbf{P}_{pp} - \mathbf{P}_{pg} \mathbf{P}_{gg}^{-1} \mathbf{P}_{gp}. \quad (15)$$

The phase-domain per-unit-length matrices follow as

$$\mathbf{R} = \text{Re}\{\mathbf{Z}_{\text{ph}}\}, \quad \mathbf{L} = \frac{1}{\omega} \text{Im}\{\mathbf{Z}_{\text{ph}}\} \cdot 10^3, \quad \mathbf{C} = \mathbf{P}_{\text{ph}}^{-1} \cdot 10^{12}, \quad \mathbf{G} = g_0 \mathbf{I}_3, \quad (16)$$

with g_0 a small leakage conductance modelling corona/insulator losses (default 2×10^{-3} $\mu\text{S}/\text{km}$). For non-transposed geometries $\mathbf{Z}_{\text{ph}}$ and $\mathbf{P}_{\text{ph}}$ are symmetric but *not* circulant; their unequal entries are the structural origin of every unbalance phenomenon that follows.

4 Distributed-Parameter Line Equations

Let $\mathbf{V}(s)$ and $\mathbf{I}(s)$ denote the phasor voltage and current vectors at distance $s \in [0, \ell]$ from the local terminal. Form the per-unit-length series impedance and shunt admittance

$$\mathbf{Z} = \mathbf{R} + j\omega \mathbf{L}, \quad \mathbf{Y} = \mathbf{G} + j\omega \mathbf{C}. \quad (17)$$

The Telegrapher’s equations in the frequency domain are

$$\frac{d}{ds} \begin{bmatrix} \mathbf{V} \\ \mathbf{I} \end{bmatrix} = \underbrace{\begin{bmatrix} \mathbf{0} & -\mathbf{Z} \\ -\mathbf{Y} & \mathbf{0} \end{bmatrix}}_{\mathbf{F}} \begin{bmatrix} \mathbf{V} \\ \mathbf{I} \end{bmatrix}. \quad (18)$$

The state vector at any distance follows from the matrix exponential

$$\phi(s) \equiv \begin{bmatrix} \mathbf{V}(s) \\ \mathbf{I}(s) \end{bmatrix} = e^{\mathbf{F}s} \phi(0). \quad (19)$$

The end-to-end ABCD transfer is the 6×6 matrix $\mathbf{M} = e^{\mathbf{F}\ell}$, partitioned as

$$\mathbf{M} = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{A}' & \mathbf{D} \end{bmatrix}, \quad \begin{bmatrix} \mathbf{V}_r \\ \mathbf{I}_r \end{bmatrix} = \mathbf{M} \begin{bmatrix} \mathbf{V}_s \\ \mathbf{I}_s \end{bmatrix}, \quad (20)$$

with subscripts s and r denoting the sending end ($s = 0$) and the receiving end ($s = \ell$).

5 Boundary Conditions: Symmetric Voltage and Specified S

Two physical constraints close the boundary problem:

1. **Balanced symmetric positive-sequence voltage at the local terminal.** The local bus is the angular reference and is held at the nominal line-to-neutral magnitude $V_s = V_{LL}/\sqrt{3}$:

$$\mathbf{V}_s = V_s \mathbf{u}. \quad (21)$$

2. **Specified three-phase complex power at the local terminal.**

$$S = P + jQ = \mathbf{V}_s^\top \mathbf{I}_s^*. \quad (22)$$

(An entirely symmetric formulation applies if the constraint is given at the remote terminal: swap $s \leftrightarrow r$, invert $\mathbf{M}$, and proceed identically.)

The remote bus is assumed sufficiently stiff to remain balanced in magnitude and angle, so

$$\mathbf{V}_r = V_r e^{j\delta} \mathbf{u}, \quad (23)$$

where V_r and δ are two unknown real scalars. Define the single complex unknown

$$\boxed{W \equiv V_r e^{-j\delta} \iff V_r = |W|, \quad \delta = -\arg(W),} \quad (24)$$

so that $V_r e^{j\delta} = W^*$.

6 Closed-Form Solution for W

Solving the upper block of (20) for $\mathbf{I}_s$ and substituting (21)–(23) gives

$$\mathbf{I}_s = \mathbf{B}^{-1}(\mathbf{V}_r - \mathbf{A} \mathbf{V}_s) = \mathbf{B}^{-1}(W^* \mathbf{u} - V_s \mathbf{A} \mathbf{u}). \quad (25)$$

Plugging (25) into (22) yields a single scalar complex equation linear in W . Expanding and collecting terms,

$$S = V_s W \underbrace{\mathbf{u} \cdot \overline{\mathbf{B}^{-1} \mathbf{u}}}_{\alpha} - V_s^2 \underbrace{\mathbf{u} \cdot \overline{\mathbf{B}^{-1} \mathbf{A} \mathbf{u}}}_{\beta}, \quad (26)$$

where the dot products use ordinary (non-Hermitian) component sums and the overline denotes complex conjugation. Both $\alpha, \beta \in \mathbb{C}$ depend only on the line (through $\mathbf{A}$ and $\mathbf{B}$), so

$$\boxed{W = \frac{S + V_s^2 \beta}{V_s \alpha}, \quad V_r = |W|, \quad \delta = -\arg(W).} \quad (27)$$

Once W is known, (25) delivers $\mathbf{I}_s$; the receiving-end voltage and current follow from (23) and (20).

7 Analytical Form of the Phase-Current Magnitudes

Substituting $\mathbf{V}_s = V_s \mathbf{u}$ and $\mathbf{V}_r = W^* \mathbf{u}$ in (25) and extracting the k -th component yields a remarkably compact expression. Introduce the two complex 3-vectors

$$\mathbf{p} := \mathbf{B}^{-1} \mathbf{u}, \quad \mathbf{q} := \mathbf{B}^{-1} \mathbf{A} \mathbf{u}, \quad (28)$$

both of which depend on the line geometry only (through $\mathbf{M} = \mathbf{e}^{\mathbf{F}\ell}$) and are independent of W and S . Then

$$I_k = W^* p_k - V_s q_k, \quad k \in \{A, B, C\}. \quad (29)$$

7.1 Closed Form for $|I_k|$

Taking modulus squared of (29) and using $W = V_r e^{-j\delta}$,

$$|I_k|^2 = V_r^2 |p_k|^2 + V_s^2 |q_k|^2 - 2 V_s V_r |p_k| |q_k| \cos(\delta + \psi_k), \quad (30)$$

where the ψ_k are line-only phase angles

$$\psi_k := \arg(p_k) - \arg(q_k), \quad k \in \{A, B, C\}. \quad (31)$$

Equation (30) is the law of cosines: each $|I_k|$ is the length of the third side of a triangle with sides $V_r |p_k|$ and $V_s |q_k|$ enclosing the angle $(\delta + \psi_k)$.

7.2 Largest Modular Deviation Between Phases

The peak-to-peak spread of the three phase magnitudes is, in closed form,

$$\Delta_{\max}(W) = \max_k \sqrt{V_r^2 |p_k|^2 + V_s^2 |q_k|^2 - 2 V_s V_r |p_k| |q_k| \cos(\delta + \psi_k)} - \min_k \sqrt{(\text{same})}. \quad (32)$$

Equivalently, every pairwise modulus difference reads

$$|I_i|^2 - |I_j|^2 = V_r^2 (|p_i|^2 - |p_j|^2) + V_s^2 (|q_i|^2 - |q_j|^2) - 2 V_s V_r [|p_i| |q_i| \cos(\delta + \psi_i) - |p_j| |q_j| \cos(\delta + \psi_j)], \quad (33)$$

which exposes the three sources of asymmetry:

- the spread of the moduli $|p_k|$, weighted by V_r^2 (receiving-side, shunt-driven asymmetry);
- the spread of the moduli $|q_k|$, weighted by V_s^2 (sending-side, series-driven asymmetry);
- a cross term modulated by $\cos(\delta + \psi_k)$, which is the only piece that depends on the transmission angle δ and hence is the only piece sensitive to the loading direction.

The relative deviation reported by the application (eq. 36) follows by normalising (30) by the mean of the three magnitudes; the maximum percentage deviation is therefore an explicit function of W alone, given the six line constants $\{|p_k|, |q_k|, \psi_k\}_{k \in \{A, B, C\}}$.

8 Voltage and Current Profile Along the Line

With the initial state $\phi(0) = [\mathbf{V}_s; \mathbf{I}_s]$ determined, the entire profile is recovered from (19):

$$\phi(s_k) = \mathbf{e}^{\mathbf{F}s_k} \phi(0), \quad s_k \in \{0, \Delta s, \dots, \ell\}. \quad (34)$$

Each phase voltage and phase current is plotted as a function of s . The receiving-end power is recomputed by direct dot product $S_r = \mathbf{V}_r^\top \mathbf{I}_r^*$, and the total active loss is $P_{\text{loss}} = P - \text{Re}\{S_r\}$.

9 Quantifying Phase-Current Unbalance

9.1 Per-Phase Magnitude and Angular-Spacing Deviations

At a chosen terminal, define magnitudes $|I_k|$ and angles $\theta_k = \arg(I_k)$ for $k \in \{A, B, C\}$, the mean magnitude $\overline{|I|} = \frac{1}{3}(|I_A| + |I_B| + |I_C|)$, and the three signed inter-phase angular spacings wrapped to $(-\pi, \pi]$:

$$s_{AB} = \text{wrap}(\theta_B - \theta_A), \quad s_{BC} = \text{wrap}(\theta_C - \theta_B), \quad s_{CA} = \text{wrap}(\theta_A - \theta_C), \quad (35)$$

with mean absolute spacing $\overline{|s|} = \frac{1}{3}(|s_{AB}| + |s_{BC}| + |s_{CA}|)$. The per-phase deviations reported are

$$\Delta_{\text{mag},k}[\%] = \frac{|I_k| - \overline{|I|}}{\overline{|I|}} \cdot 100, \quad \Delta_{\text{ang},k}[^{\circ}] = |s_k| - \overline{|s|}. \quad (36)$$

9.2 Spatial Variation Along the Line

Because the per-unit-length matrices are non-circulant, each phase current magnitude is not constant along s :

$$\eta_k[\%] = \frac{\max_s |I_k(s)| - \min_s |I_k(s)|}{\min_s |I_k(s)|} \cdot 100, \quad k \in \{A, B, C\}. \quad (37)$$

9.3 Relation to the Standard Current Unbalance Factor

For comparison with international standards, the symmetrical-component decomposition

$$\begin{bmatrix} I_0 \\ I_+ \\ I_- \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} I_A \\ I_B \\ I_C \end{bmatrix} \quad (38)$$

yields the current unbalance factor

$$\text{IUF} = \frac{|I_-|}{|I_+|} \cdot 100 [\%]. \quad (39)$$

10 Unbalance as a Function of Length and Apparent Power

Equations (27)–(25) make the dependence of the solution on ℓ and S explicit. The line-dependent quantities $\mathbf{A}(\ell)$ and $\mathbf{B}(\ell)$ come exclusively from $\mathbf{M} = \mathbf{e}^{\mathbf{F}\ell}$, so:

- In the electrically short limit $\mathbf{A} \rightarrow \mathbf{I}_3$ and $\mathbf{B} \rightarrow \ell \mathbf{Z}$, so the sending current reduces to $\mathbf{I}_s \approx (\ell \mathbf{Z})^{-1}(\mathbf{W}^* \mathbf{u} - V_s \mathbf{u})$: the dominant asymmetry is that of $\mathbf{Z}$ itself, scaled by $1/\ell$.
- As ℓ grows the shunt admittance contributes through $\mathbf{B}^{-1} \mathbf{A}$; the asymmetry inherited from the non-circulant geometry propagates into both α and β .
- For fixed line geometry, $S = P + jQ$ sets the absolute magnitude of $\mathbf{I}_s$ but only weakly modifies the relative unbalance (36)–(37), because the constant capacitive term $V_s^2 \beta$ becomes a smaller fraction of the loading term as $|S|$ grows. Light loading therefore typically produces a larger percentage unbalance than heavy loading.

The Litgrid application sweeps (ℓ, S) by re-evaluating (27), (25) and (36)–(37) for each pair.

11 Worked Example: 110 kV Single-Circuit Line

11.1 Inputs

A horizontal-vertical asymmetric configuration with one conductor per phase and one overhead ground wire.

- Nominal line-to-line voltage $V_{LL} = 110$ kV, so the local line-to-neutral magnitude is $V_s = 110/\sqrt{3} \approx 63\,509.1$ V.
- Three-phase complex power injected at the local terminal: $S = P + jQ = 80 + j10$ MV A.
- Frequency $f = 50$ Hz, so $\omega = 100\pi \approx 314.159$ rad/s.
- Soil resistivity $\rho = 1000$ Ω m.
- Phase conductors (one per phase, no bundling): outer diameter 17.1 mm (so $r_p = 8.55$ mm), $r_{dc} = 0.1897$ Ω /km.
- Ground wire: outer diameter 11.11 mm ($r_g = 5.555$ mm), $r_{dc} = 2.53$ Ω /km.
- Geometry (order A, B, C, G): $x = (-4, 0, +4, 0)$ m and $y = (20, 20, 23, 25)$ m.

There is one conductor per phase, so the bundle reduction is the identity ($\mathbf{T} = \mathbf{I}_4$). Only the ground-wire elimination is non-trivial.

11.2 Geometric Distances and Image Distances

Real distances d_{ij} [m] and image distances D_{ij} [m]:

pair	$A-B$	$B-C$	$A-C$	$A-G$
d_{ij}	4.000	5.000	$\sqrt{73} \approx 8.544$	$\sqrt{41} \approx 6.403$
D_{ij}	$\sqrt{1616} \approx 40.199$	$\sqrt{1865} \approx 43.186$	$\sqrt{1913} \approx 43.738$	$\sqrt{2041} \approx 45.177$
pair	$B-G$	$C-G$		
d_{ij}	5.000	$\sqrt{20} \approx 4.472$		
D_{ij}	45.000	$\sqrt{2320} \approx 48.166$		

The self-image distances are $2y_A = 2y_B = 40$, $2y_C = 46$, $2y_G = 50$ m.

11.3 Carson Constants

$$\begin{aligned}
 D_e &= 658.5 \sqrt{1000/50} \approx 2944.65 \text{ m}, \\
 R_e &= \frac{\omega \mu_0}{8\pi} \cdot 10^3 \approx 0.015\,71 \text{ } \Omega/\text{km}, \\
 \frac{\omega \mu_0}{2\pi} \cdot 10^3 &\approx 0.062\,83 \text{ } \Omega/\text{km per unit ln}.
 \end{aligned}$$

Note that numerically the internal reactance per kilometre equals the earth-return resistance: $\omega L_{\text{int}} \cdot 10^3 = R_e = 0.015\,71$ Ω /km.

11.4 Coupled 4×4 Series Impedance Matrix

Apply (6)–(7). For instance,

$$\begin{aligned} Z_{AA} &= (0.1897 + 0.01571) + j[0.01571 + 0.06283 \ln(2944.65/0.00855)] \\ &= 0.2054 + j0.8167 \text{ } [\Omega/\text{km}], \\ Z_{AB} &= 0.01571 + j0.06283 \ln(2944.65/4) = 0.01571 + j0.4149, \\ Z_{GG} &= (2.53 + 0.01571) + j[0.01571 + 0.06283 \ln(2944.65/0.005555)] \\ &= 2.5457 + j0.8438. \end{aligned}$$

Repeating for every pair, the fully coupled cable-to-cable matrix $\mathbf{Z}_{\text{full}}$, in Ω/km , with rows/columns ordered (A, B, C, G) , is

$$\mathbf{Z}_{\text{full}} = \begin{bmatrix} 0.2054+j0.8167 & 0.0157+j0.4149 & 0.0157+j0.3671 & 0.0157+j0.3852 \\ 0.0157+j0.4149 & 0.2054+j0.8167 & 0.0157+j0.4008 & 0.0157+j0.4008 \\ 0.0157+j0.3671 & 0.0157+j0.4008 & 0.2054+j0.8167 & 0.0157+j0.4078 \\ 0.0157+j0.3852 & 0.0157+j0.4008 & 0.0157+j0.4078 & 2.5457+j0.8438 \end{bmatrix}.$$

11.5 Kron Elimination of the Ground Wire

With $\mathbf{T} = \mathbf{I}_4$, the bundle-reduced matrix coincides with $\mathbf{Z}_{\text{full}}$. Partitioning the last row/column as the ground block and applying (14) produces the 3×3 phase-domain matrix

$$\mathbf{Z}_{\text{ph}} = \begin{bmatrix} 0.2564+j0.7951 & 0.0688+j0.3923 & 0.0698+j0.3443 \\ 0.0688+j0.3923 & 0.2607+j0.7935 & 0.0720+j0.3771 \\ 0.0698+j0.3443 & 0.0720+j0.3771 & 0.2627+j0.7928 \end{bmatrix} \text{ } [\Omega/\text{km}].$$

Separating real and imaginary parts according to (16):

$$\mathbf{R} = \begin{bmatrix} 0.2564 & 0.0688 & 0.0698 \\ 0.0688 & 0.2607 & 0.0720 \\ 0.0698 & 0.0720 & 0.2627 \end{bmatrix} \Omega/\text{km}, \quad \mathbf{L} = \begin{bmatrix} 2.531 & 1.249 & 1.096 \\ 1.249 & 2.526 & 1.200 \\ 1.096 & 1.200 & 2.524 \end{bmatrix} \text{mH/km}.$$

Note that the three diagonal inductances differ by less than 0.5% but the three off-diagonals span 1.10 to 1.25 mH/km: this asymmetry, induced by the non-symmetric arrangement of phase C relative to the A/B row, is what drives the eventual current unbalance.

11.6 Potential-Coefficient Matrix and Phase Capacitance

With $1/(2\pi\epsilon_0) \approx 1.7975 \times 10^{10}$ m/F, (8)–(9) give, in units of 10^{10} m/F,

$$\mathbf{P}_{\text{full}} = \begin{bmatrix} 15.191 & 4.148 & 2.935 & 3.513 \\ 4.148 & 15.191 & 3.875 & 3.950 \\ 2.935 & 3.875 & 15.443 & 4.272 \\ 3.513 & 3.950 & 4.272 & 16.366 \end{bmatrix}.$$

Applying (15) to eliminate the ground row/column,

$$\mathbf{P}_{\text{ph}} = \begin{bmatrix} 14.437 & 3.300 & 2.018 \\ 3.300 & 14.238 & 2.844 \\ 2.018 & 2.844 & 14.328 \end{bmatrix} [10^{10} \text{ m/F}],$$

and the phase capacitance matrix $\mathbf{C} = \mathbf{P}_{\text{ph}}^{-1}$ becomes, in nF/km,

$$\mathbf{C} = \begin{bmatrix} 7.387 & -1.567 & -0.730 \\ -1.567 & 7.646 & -1.297 \\ -0.730 & -1.297 & 7.341 \end{bmatrix}.$$

Again the entries are visibly non-circulant.

11.7 Per-Unit-Length $\mathbf{Z}$ and $\mathbf{Y}$

At $\omega = 100\pi$:

$$\mathbf{Z} = \mathbf{R} + j\omega\mathbf{L} \cdot 10^{-3}, \quad \mathbf{Y} = g_0\mathbf{I}_3 \cdot 10^{-6} + j\omega\mathbf{C} \cdot 10^{-9},$$

both expressed in Ω/km and S/km respectively. $\mathbf{Z}$ coincides with $\mathbf{Z}_{\text{ph}}$ above, and $\mathbf{Y}$ has diagonal entries $\approx j 2.32 \times 10^{-6} \text{ S}/\text{km}$ with off-diagonals roughly half a microsiemens per kilometre.

11.8 Sending-End Solution at Selected Line Lengths

Compute $\mathbf{M} = \mathbf{e}^{F\ell}$ numerically, take $\mathbf{A}$, $\mathbf{B}$ from its upper blocks, evaluate $\alpha = \mathbf{u} \cdot \overline{\mathbf{B}^{-1}\mathbf{u}}$ and $\beta = \mathbf{u} \cdot \overline{\mathbf{B}^{-1}\mathbf{A}\mathbf{u}}$, and obtain W from (27). The sending current $\mathbf{I}_s$ follows from (25). The table below collects the result for $S = 80 + j10 \text{ MV A}$ at four representative lengths:

ℓ [km]	$ I_A $ [A]	$ I_B $ [A]	$ I_C $ [A]	Δ_A [%]	Δ_B [%]	Δ_C [%]	V_r [kV LN]
40	407.67	446.21	417.07	-3.77	+5.33	-1.55	59.74
80	407.59	446.31	417.06	-3.79	+5.35	-1.56	56.69
120	407.51	446.41	417.05	-3.81	+5.37	-1.56	54.49
200	407.39	446.58	417.02	-3.84	+5.41	-1.57	53.04

The mean sending current magnitude is $|\overline{I}| = (P^2 + Q^2)^{1/2}/(\sqrt{3} V_{LL}) \approx 423.3 \text{ A}$, in good agreement with the row averages above.

11.9 Interpretation

- Phase B carries roughly 5 % more current than the three-phase mean, while phase A carries 3.8 % less and phase C about 1.6 % less. The middle phase being the most loaded is a direct consequence of the geometry: phase B sits at the lowest height, equidistant from A and C , and therefore carries the largest share of the imbalanced charging current dictated by the non-circulant $\mathbf{C}$ and $\mathbf{L}$ matrices computed above.
- The percentage deviations are remarkably stable over the range 40–200 km: the structural asymmetry of $\mathbf{Z}$ and $\mathbf{Y}$ sets the unbalance pattern, not the line length. Length affects mainly the absolute magnitudes (slight bias because the capacitive charging current adds to the loading current) and the receiving-end voltage and angle.
- The receiving-end voltage drops monotonically with ℓ , from 59.7 kV at 40 km (a 5.9 % drop relative to $V_s = 63.5 \text{ kV}$) down to 53.0 kV at 200 km (16.5 % drop), reflecting both active losses across $\mathbf{R}$ and reactive drop across $\mathbf{L}$.
- Sweeping S at fixed ℓ reveals the second axis of the $\Delta_{\text{mag},k}(\ell, S)$ surface: as $|S|$ decreases the capacitive term $V_s^2\beta$ in (27) dominates over the loading term S , and the percentage unbalance grows in magnitude because the fixed shunt-induced charging current becomes a larger fraction of $|\mathbf{I}_s|$.

The full surface $\Delta_{\text{mag},k}(\ell, S)$ and the spatial-variation indicator $\eta_k(\ell, S)$ of (37) are produced by the Litgrid application by repeating the procedure above on a parameter grid.